

ADAPTIVE CONFIDENCE AND FEATURE STABILITY GUIDED EARLY EXIT NETWORKS FOR EFFICIENT IMAGE RECOGNITION

Research Study Report

Task Date: 20/07/2026

1. Introduction and Motivation

Modern deep learning models for image recognition have grown increasingly deeper and more accurate over the past decade — from AlexNet with 8 layers to ResNet with 152 layers. However, this progression toward depth has come at a significant cost: every image, whether it is a simple photograph of a clear sky or a blurry, partially occluded scene, passes through the exact same number of layers in the network. This is fundamentally wasteful.

Consider how a human expert makes decisions. A cardiologist does not spend the same amount of time examining an ECG showing a perfectly normal heartbeat as one showing an irregular pattern requiring careful analysis. The easy case is recognized quickly; the difficult case earns deeper attention. Deep neural networks currently cannot make this distinction — they treat all inputs with equal computational effort regardless of difficulty.

This is the core problem our research addresses. The goal is to build a CNN-based image recognition system that can assess its own certainty and decide, at each layer of processing, whether it has gathered enough information to make a confident, stable prediction — or whether it needs to continue processing through deeper layers.

The question the system should ask at each intermediate layer is not simply "how confident am I?" but the more meaningful question: "If I were to process this image through two more layers, would my prediction change?" If the answer is almost certainly no — exit early. If there is a real risk of the prediction shifting, continue.

2. Problem Statement

The existing approach to early-exit neural networks relies predominantly on a single signal to determine when to stop processing: the softmax probability output. If the softmax confidence score exceeds a fixed threshold, the sample exits. This approach suffers from a well-documented and serious problem — softmax probabilities are poorly calibrated.

Research has demonstrated that a model can output a softmax probability of 99% and still produce an incorrect classification. This happens because the softmax function is sensitive to the relative magnitudes

of logit scores rather than to any absolute notion of correctness. A model that has learned to produce very large logit differences will appear highly confident even on inputs it has never seen.

The central problem: No existing method for image classification combines multiple independent signals, including how much the prediction is changing between layers and how much the internal feature representations are shifting into a unified, trainable exit decision module. Single-signal methods are individually weak. The question is whether a principled combination of multiple signals can make exit decisions substantially more reliable.

3. What Already Exists:

A thorough review of the existing literature reveals five categories of prior work, each addressing early exit from a different angle. Understanding these approaches — and their individual limitations — is what motivates our proposed direction.

3.1 BranchyNet (2017)

BranchyNet introduced the foundational architecture for early-exit networks. The core idea is to add side branch classifiers at intermediate points in a deep network. During inference, if the entropy of the prediction at any branch falls below a learned threshold, the sample exits at that branch without passing through the remaining layers.

Key results: BranchyNet achieved 2x–6x speedup on CPU and GPU for LeNet, AlexNet, and ResNet on MNIST and CIFAR-10. For B-LeNet, 94.3% of samples exited at the first branch with 5.4x CPU speedup while maintaining accuracy.

Exit criterion: Entropy of the softmax output only.

Core limitation: The paper itself acknowledges that entropy thresholds are manually set and sensitive to dataset difficulty. There is no mechanism for the network to assess whether its representation has stabilized. The method gives no signal about whether processing additional layers would change the prediction.

3.2 Shallow-Deep Networks (2019)

This paper introduced and formally characterized the concept of "overthinking" in deep neural networks — a situation where a network arrives at the correct classification at an intermediate layer but then, after further processing, changes its mind and produces an incorrect final prediction. The paper showed this occurs for 50% of misclassifications on natural inputs.

Exit criterion: Softmax confidence score (maximum predicted probability).

Key insight: Overthinking is not just wasteful — it is actively harmful. Additional computation can corrupt a correct answer. This provides strong motivation for any early-exit system to be aware of prediction stability, not just confidence.

Core limitation: Still uses a single score (softmax max probability) that is known to be overconfident. Does not track how the prediction is evolving across layers.

3.3 MSDNet — Multi-Scale Dense Networks (2018)

MSDNet addressed an architectural limitation of naive early-exit networks: the fact that early layers in standard CNNs lack coarse-scale features, which are critical for classification. The method uses dense connectivity and multi-scale feature extraction to ensure that even early classifiers have access to representations of multiple spatial scales.

Exit criterion: Confidence threshold on softmax probabilities.

Key insight: Exit head placement and architecture significantly affect early classifier quality. Poor early classifiers will cause the exit criterion to fail regardless of how sophisticated the threshold is.

Core limitation: The exit decision mechanism itself remains confidence-based only. The architectural improvements in MSDNet and the exit criterion design are treated as separate concerns.

3.4 DeeBERT (2020)

DeeBERT adapted early-exit strategies to large transformer-based language models, demonstrating up to 40% inference time reduction with minimal accuracy loss. The approach adds off-ramp classifiers after each transformer layer and uses the entropy of the prediction distribution as the exit signal.

Exit criterion: Entropy alone.

Core limitation: Confidence (entropy) of a prediction at a single layer says nothing about whether that confidence is stable or whether deeper layers would produce a different result. The paper itself notes that entropy can be misleadingly low even when the prediction is about to change.

3.5 PABEE — Patience-Based Early Exit (2020)

PABEE introduced prediction stability as an exit criterion. The model exits when the predicted class remains the same across t consecutive internal classifiers, where t is a pre-defined "patience" hyperparameter. This was a meaningful advance: instead of asking "how confident is the model right now?", it asks "has the model been consistently predicting the same class for multiple layers?"

Exit criterion: Cross-layer prediction consistency — exit when $\text{argmax}(y_i) == \text{argmax}(y_{i-1})$ for t consecutive layers.

Key result: Outperformed entropy-only methods by a significant margin on GLUE benchmark tasks, confirming that prediction stability carries signal that softmax entropy does not.

Core limitation: The stability signal is purely binary (same class or not). It does not measure the degree of stability across the full probability distribution, and it completely ignores how much the internal feature representations have changed. A model could predict the same top class for three layers while having very different underlying probability distributions and feature maps.

3.6 PCEE-BERT (2022)

PCEE-BERT combined two signals — entropy-based confidence and patience-based prediction consistency — into a hybrid exit criterion. An input exits early if both conditions are satisfied: the entropy is below a threshold AND the prediction has been consistent for a required number of consecutive layers.

Exit criterion: Entropy AND prediction consistency combined.

Key result: Consistently outperformed PABEE, especially when high speedup ratios are required, confirming that combining two signals is better than one.

Core limitation: Still uses only two signals out of a possible richer set. Feature-level stability — how much the internal representations are changing — is not considered. The combination is also fixed: both signals get equal logical weight (both must be satisfied), which does not allow the model to learn which signal matters more for different types of inputs.

3.7 COSEE (2025)

The most recent and closely related work, COSEE introduced a consistency-oriented exit framework with calibrated sample weighting. It addresses the inconsistency between training and testing behaviors in early-exit models, arguing that easy and hard samples should not be treated equally during training.

Exit criterion: Normalized energy score with online signal calibration.

Core limitation: Though more sophisticated in training design, the exit signal itself remains energy/confidence-based. Feature stability and margin between top predictions are not incorporated.

4. Identified Gaps : Where Existing Work Falls Short

After reviewing the literature, the following gaps emerge clearly:

Method	Signals Used for Exit Decision	What Is Missing
BranchyNet	Entropy	Stability signals; learned threshold
Shallow-Deep Networks	Softmax max probability	Any stability signal
MSDNet	Softmax confidence	Any stability signal
DeeBERT	Entropy	Stability signals
PABEE	Prediction consistency only	Confidence + feature-level signal
PCEE-BERT	Entropy + prediction consistency	Feature stability; learned weighting
COSEE	Energy score	Stability signals; margin

Three structural gaps emerge across all existing methods:

Gap 1 — No method combines all four signals: Every existing method uses at most two of the four relevant signals (confidence/entropy, prediction stability, feature stability, margin). No method has explored what happens when all four are considered together.

Gap 2 — All combination weights are fixed: Where multiple signals are used (e.g., PCEE-BERT), the combination is hardcoded — either a logical AND or a fixed linear sum with pre-set coefficients. No existing method learns which signal should dominate for a given input or a given dataset. For an easy, high-contrast image, entropy alone might be sufficient. For a partially occluded or blurry image, feature stability might matter far more. A fixed weighting ignores this input-dependent variation.

Gap 3 — Feature stability is entirely absent from image classification: PABEE and PCEE-BERT use prediction stability in the NLP domain, but feature stability; measuring how much the internal feature representations change between consecutive layers; has not been incorporated into any early-exit criterion for CNN-based image classification. This is a notable omission, because if the feature maps have stopped changing substantially, the network has almost certainly converged on a stable representation and further processing is unlikely to change the final output.

5. Our Proposed Approach

5.1 Core Idea

We propose replacing the single-signal or two-signal exit criteria of existing methods with a composite exit score computed from four independent signals, with the combination weights themselves learned during training. The exit decision at each intermediate layer is governed by:

$$\text{ExitScore} = \alpha \cdot (1 - \text{Entropy}) + \beta \cdot (\text{Prediction Stability}) + \gamma \cdot (\text{Feature Stability}) + \delta \cdot (\text{Margin})$$

Where α , β , γ , δ are not manually set hyperparameters but parameters learned during training, and the four signals are computed as follows.

5.2 The Four Signals Explained

Signal	How It Is Computed and What It Means
1 – Entropy (Confidence)	Entropy = $-\sum p \cdot \log(p)$ over all classes. Low entropy means the model assigns most probability to one class. We use (1–entropy) so higher values mean higher confidence.
Prediction Stability	Cosine similarity or KL divergence between the probability vectors at layer L and layer L–1. If the vectors are very similar, the prediction is not changing with additional processing. High similarity means stable.
Feature Stability	$\ F_L - F_{L-1}\ / \ F_{L-1}\ $, where F_L is the feature map (internal representation) at layer L, normalized by the previous layer's norm. If the feature map is barely changing, the network has likely extracted all available information.
Margin	$P(\text{top class}) - P(\text{second class})$. The gap between the top two predicted probabilities. A large margin means the model strongly prefers one class over all others, which is a different and complementary signal to entropy.

5.3 Our Novel Contribution — Learned Adaptive Weights

The single most important novel element of our proposal is that the weights α , β , γ , δ are not fixed by hand. Instead, they are learned parameters optimized during training as part of a small gating sub-network. This sub-network takes the current layer index and, optionally, a compressed representation of the input features, and outputs the four weights for that specific input.

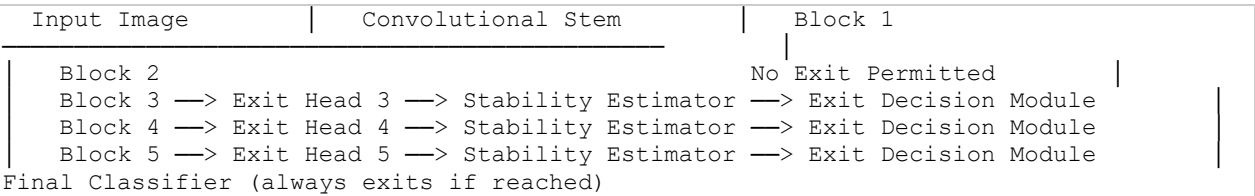
This is motivated by a straightforward observation: different types of images and different types of errors call for different signals. A clear, well-lit photograph of a single object on a plain background is best handled by confidence (entropy) alone — it converges quickly and the confidence signal is reliable. A cluttered, ambiguous scene where multiple objects are present might require patience for prediction stability before exiting. An image degraded by blur or occlusion might need feature stability as the primary criterion, because the model's representations keep shifting as it processes more spatial context.

By learning these weights from data, the system can adapt its exit policy to the dataset and the difficulty distribution of inputs — something no existing method attempts.

6. Proposed Architecture

6.1 Overall Network Structure

The backbone CNN (initially ResNet-50) is modified to include exit heads at every major block. The number of blocks before the first permitted exit is configurable; we propose a minimum of 2 blocks to ensure the model has learned some meaningful representation before attempting early classification.



6.2 Components

Exit Head

A lightweight classifier (global average pooling → linear layer → softmax) attached after each block. It produces the probability vector used for confidence, prediction stability, and margin computation.

Stability Estimator

A module that compares the current block's outputs (both prediction vector and feature map) with those stored from the previous block. It outputs four scalar values — the four signals described above — and passes them to the Exit Decision Module.

Exit Decision Module

A small MLP (2 layers, ~64 hidden units) that takes the four signals as input and outputs a single scalar exit score. The weights of this MLP are what we train to be input-adaptive. At inference time, if this score exceeds a threshold, the sample exits; otherwise it continues to the next block.

Why MLP instead of fixed formula?

A fixed formula like $\alpha(1-\text{Entropy}) + \beta(\text{Prediction Stability}) + \gamma(\text{Feature Stability}) + \delta(\text{Margin})$ requires the researcher to decide how to weight the signals. Using a small trained MLP allows the model to learn non-linear relationships between the signals and the optimal exit decision. A reviewer would rightly ask whether a fixed formula was the best combination; a learned module sidesteps this concern entirely by letting the data decide.

7. Training Strategy

Training early-exit networks is non-trivial because all exit heads must be trained simultaneously. If early exits dominate the loss, deep layers receive little gradient signal and do not learn. If deep exits dominate, early exits never become accurate enough to be useful.

7.1 Joint Loss Formulation

Following the BranchyNet convention, we use a weighted sum of cross-entropy losses across all exit points:

$$L_{\text{total}} = \sum_k w_k \cdot \text{CrossEntropy}(y_{\text{hat}_k}, y_{\text{true}})$$

Where k indexes exit points, and w_k are weights that give more emphasis to earlier exits to ensure they learn discriminative features. Based on BranchyNet's findings, we will start with higher weights for earlier exits (e.g., $w = [1.0, 0.8, 0.6, 0.4, 0.2]$) and tune based on validation performance.

7.2 Exit Decision Module Training

The Exit Decision Module is trained end-to-end with the backbone. It receives a supervision signal derived from oracle exit labels — for each training sample, we compute which exit point would have given the correct classification with minimum computation, and train the exit module to predict this oracle decision. This provides direct supervision for the exit decision rather than relying solely on the accuracy gradient.

8. Datasets and Evaluation Plan

8.1 Datasets

Dataset	Purpose and Justification
CIFAR-10	Standard baseline — 10 easy classes, used in BranchyNet and SDN for direct comparison
CIFAR-100	Higher difficulty — 100 classes require finer discrimination, tests whether feature stability helps
Tiny ImageNet	Larger scale — 200 classes, 64×64 images; bridges gap between CIFAR and full ImageNet
CUB-200-2011 (Birds)	Fine-grained recognition — small inter-class differences; hardest test for exit decisions

Stanford Cars	Fine-grained recognition — another domain to test generalization of learned exit weights
---------------	--

8.2 Evaluation Metrics

Beyond top-1 accuracy, we will report the following metrics for a complete evaluation:

- Average FLOPs per sample (primary efficiency metric)
- Average inference latency (wall-clock time, batch size 1)
- Percentage of samples exiting at each block
- Expected Calibration Error (ECE) — to measure whether our exit scores are well-calibrated
- Accuracy-FLOPs trade-off curve — the primary comparison curve against baselines

9. Ablation Study Design

The ablation study is what will establish that each component of our approach contributes independently. We will train the following configurations:

Configuration	Exit Criterion	What It Tests
Baseline-E	Entropy only	Replication of BranchyNet
Baseline-P	Prediction stability only	Replication of PABEE idea
Baseline-M	Margin only	Novel single-signal baseline
Baseline-F	Feature stability only	Novel single-signal baseline
Combo-2A	Entropy + Prediction Stability (fixed)	Replication of PCEE-BERT idea
Combo-2B	Entropy + Feature Stability (fixed)	Novel two-signal baseline
Combo-3	Entropy + Pred. Stability + Feature Stability (fixed)	Three-signal, fixed weights
Full-Fixed	All 4 signals, fixed $\alpha \beta \gamma \delta$	Four-signal, no learned weighting
Full-Learned (Ours)	All 4 signals, learned Exit Decision MLP	Our full proposed method

This progression allows us to attribute performance gains to (1) adding more signals, and (2) making the combination learned rather than fixed.

10. Summary of Novel Contributions

When presenting this work, the following contributions can be clearly stated:

Contribution 1 — Multi-signal exit criterion for image classification: We are the first to combine all four signals (confidence, prediction stability, feature stability, margin) into a single exit decision for CNN-based image recognition. Prior work used at most two signals, and feature stability has not previously been used in any image classification early-exit method.

Contribution 2 — Learned adaptive weighting: Instead of fixing the combination weights manually, we train a small neural module that learns to weight the four signals based on input characteristics and training data. This eliminates a critical hyperparameter and allows the exit policy to adapt to the difficulty distribution of a dataset.

Contribution 3 — Systematic ablation across signal combinations: By comparing all single-signal, two-signal, three-signal, and four-signal configurations, we provide the most comprehensive study to date of which signals contribute what value to early exit decisions in image recognition.

Contribution 4 — Evaluation on fine-grained datasets: We evaluate on CUB-200 and Stanford Cars, which are significantly harder than CIFAR-10 and represent a more meaningful test of adaptive inference. Easy datasets may not reveal the advantages of multi-signal exit criteria; fine-grained datasets expose the weaknesses of single-signal approaches.

11. References

- [1] Teerapittayanon, S., McDanel, B., Kung, H.T. (2017). BranchyNet: Fast Inference via Early Exiting from Deep Neural Networks. arXiv:1709.01686.
- [2] Kaya, Y., Hong, S., Dumitras, T. (2019). Shallow-Deep Networks: Understanding and Mitigating Network Overthinking. ICML 2019. arXiv:1810.07052.
- [3] Huang, G., Chen, D., Li, T., Wu, F., van der Maaten, L., Weinberger, K.Q. (2018). Multi-Scale Dense Networks for Resource Efficient Image Classification. ICLR 2018. arXiv:1703.09844.
- [4] Xin, J., Tang, R., Lee, J., Yu, Y., Lin, J. (2020). DeeBERT: Dynamic Early Exiting for Accelerating BERT Inference. ACL 2020. arXiv:2004.12993.
- [5] Zhou, W., Xu, C., Ge, T., McAuley, J., Xu, K., Wei, F. (2020). BERT Loses Patience: Fast and Robust Inference with Early Exit. NeurIPS 2020. arXiv:2006.04152.
- [6] Zhang, Z., Zhu, W., Zhang, J., Wang, P., Jin, R., Chung, T.S. (2022). PCEE-BERT: Accelerating BERT Inference via Patient and Confident Early Exiting. NAACL 2022.
- [7] He, J., Zhang, Q., Zhang, H., Huang, X., Naseem, U., Miao, D. (2024). COSEE: Consistency-Oriented Signal-Based Early Exiting via Calibrated Sample Weighting Mechanism. AAAI 2025. arXiv:2412.13236.